

MA10207 Analysis 1 - Semester 1, 2021/22

Problem Sheet Week 1: Solutions

1. A tautology is a statement that is true no matter what the truth values of its constituent statements are. Using the truth table, decide which of the following are tautologies.

$$(a) \quad ((P \Rightarrow Q) \vee (Q \Rightarrow P)) \wedge (\neg Q)$$

$$(b) \quad (P \Rightarrow \neg Q) \vee (Q \Rightarrow \neg R)$$

Answer. The truth table for (a) is

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$(P \Rightarrow Q) \vee (Q \Rightarrow P)$	$(\neg Q)$	$((P \Rightarrow Q) \vee (Q \Rightarrow P)) \wedge (\neg Q)$
true	true	true	true	true	false	false
true	false	false	true	true	true	true
false	true	true	false	true	false	false
false	false	true	true	true	true	true

The statement $((P \Rightarrow Q) \vee (Q \Rightarrow P)) \wedge (\neg Q)$ is not a tautology. On the other hand, observe that the statement $(P \Rightarrow Q) \vee (Q \Rightarrow P)$ is a tautology.

The truth table for (b) is

P	Q	R	$\neg Q$	$P \Rightarrow \neg Q$	$\neg R$	$Q \Rightarrow \neg R$	$(P \Rightarrow \neg Q) \vee (Q \Rightarrow \neg R)$
true	true	true	false	false	false	false	false
true	true	false	false	false	true	true	true
true	false	true	true	true	false	true	true
true	false	false	true	true	true	true	true
false	true	true	false	true	false	false	true
false	true	false	false	true	true	true	true
false	false	true	true	true	false	true	true
false	false	false	true	true	true	true	true

The statement $(P \Rightarrow \neg Q) \vee (Q \Rightarrow \neg R)$ is not a tautology.

2. Assume that the statement

$$[\neg(P \Rightarrow Q) \wedge (R \vee Q)]$$

is true. Find the truth value of P , Q and R .

Answer. Since $[\neg(P \Rightarrow Q) \wedge (R \vee Q)]$ is true, $\neg(P \Rightarrow Q)$ is true and $(R \vee Q)$ is true.

Since $\neg(P \Rightarrow Q)$ is true, $P \Rightarrow Q$ is false, or equivalently $\neg P \vee Q$ is false. This implies that both $\neg P$ and Q are false. So we have that P is true and Q is false.

Since $(R \vee Q)$ is true and Q is false, then necessarily R is true.

We conclude that P is true, Q is false and R is true.

3. Let $A = \{2, 1, -1, -3\}$ and $B = \{1, 2, 3, 4\}$, and consider the following statement

$$(\exists x \in A)(\forall y \in B)(y^2 < x - 2 \Rightarrow 2y < 2 + x).$$

Prove that this statement is true and write its negation.

Answer. We show that the statement is true and write its negation.

Take $x = -3 \in A$. If we show that $\forall y \in B : y^2 < -5 \Rightarrow 2y < -1$ is true, then the statement is true. Observe that for any $y \in B$, $y^2 < -5$ is false. Hence $\forall y \in B : y^2 < -5 \Rightarrow 2y < -1$ is true.

We conclude that the statement $(\exists x \in A)(\forall y \in B)(y^2 < x - 2 \Rightarrow 2y < 2 + x)$ is true.

Negation of the statement. $(\forall x \in A)(\exists y \in B)((y^2 < x - 2) \wedge (2y \geq 2 + x))$.

4. Let $A = \{2, 1, -1, -3\}$ and $B = \{-4, 1, -\frac{1}{2}\}$, and consider the following statements

$$(a) \quad (\forall x \in A)(\forall y \in B) (\neg(x - y > 1) \vee 2x + 3y < 2),$$

$$(b) \quad (\exists x \in A)(\forall y \in B) \left(x > 1 + \frac{y}{2} \Rightarrow \max(2x, 3y) = 1 \right).$$

Determine the truth value of each statement.

Answer. Using de Morgan's rule, we have that the negation of statement (a) is

$$\neg(a) \quad (\exists x \in A)(\exists y \in B) (x - y > 1 \wedge 2x + 3y \geq 2).$$

Statement $\neg(a)$ is true: take $x = 2 \in A$, $y = -\frac{1}{2} \in B$ and check that

$$x - y = 2 + \frac{1}{2} > 1 \quad \wedge \quad 2x + 3y = 4 - \frac{3}{2} \geq 2 \quad \text{is true.}$$

Thus statement (a) is false.

Statement (b). Take $x = -3 \in A$. Then for all $y \in B$, the statement $-3 > 1 + \frac{y}{2}$ is false: indeed we easily check that if $y = -4$, then $-3 > 1 - 2$ is false; if $y = 1$, then $-3 > 1 + \frac{1}{2}$ is false; if $y = -\frac{1}{2}$, then $-3 > 1 - \frac{1}{4}$ is false.

Hence there exists $x = -3 \in A$ such that the statement for all $y \in B$, $x > 1 + \frac{y}{2} \Rightarrow \max(2x, 3y) = 1$ is true. We conclude that statement (b) is true.

The negation of (b): $(\forall x \in A)(\exists y \in B) ((x > 1 + \frac{y}{2}) \wedge \max(2x, 3y) \neq 1)$.

Homework. Choose one between 2a and 2b.

1. Using the truth table, decide if the statements $(P \Rightarrow Q) \Rightarrow Q$ and $P \wedge Q$ are equivalent.

Answer. We consider the truth table

P	Q	$P \Rightarrow Q$	$(P \Rightarrow Q) \Rightarrow Q$	$P \wedge Q$
true	true	true	true	true
true	false	false	true	false
false	true	true	true	false
false	false	true	false	false

The two statements are not equivalent because they have different truth tables.

- 2a. Assume that $\neg P \wedge Q$ is false. Prove that

$$[(P \vee Q) \Leftrightarrow (P \wedge \neg Q)] \vee Q$$

is true.

Answer. We consider the truth table

P	Q	$\neg P \wedge Q$	$P \vee Q$	$P \wedge \neg Q$	$(P \vee Q) \Leftrightarrow (P \wedge \neg Q)$	$[(P \vee Q) \Leftrightarrow (P \wedge \neg Q)] \vee Q$
true	true	false	true	false	false	true
true	false	false	true	true	true	true
false	true	true	true	false	false	true
false	false	false	false	false	true	true

The statement $[(P \vee Q) \Leftrightarrow (P \wedge \neg Q)] \vee P$ is always true. It is a tautology.

- 2b. Assume that $\neg P \wedge Q$ is false. Prove that

$$[(P \vee Q) \Leftrightarrow (P \wedge \neg Q)] \vee P$$

is true.

Answer. We consider the truth table

P	Q	$\neg P \wedge Q$	$P \vee Q$	$P \wedge \neg Q$	$(P \vee Q) \Leftrightarrow (P \wedge \neg Q)$	$[(P \vee Q) \Leftrightarrow (P \wedge \neg Q)] \vee P$
true	true	false	true	false	false	true
true	false	false	true	true	true	true
false	true	true	true	false	false	false
false	false	false	false	false	true	true

Given that $\neg P \wedge Q$ is false, the cases to consider are: P true and Q true; P true and Q false; P false and Q false. In all these cases, the truth table gives that $[(P \vee Q) \Leftrightarrow (P \wedge \neg Q)] \vee P$ is true.

3. Let $A = \{2, 1, -1, -3\}$ and $B = \{1, 2, 3, 4\}$, and consider the following statement

$$(\forall x \in B)(\forall y \in A)(\neg(2x - y > 1) \vee (y^2 - x \text{ is a multiple of } 3))$$

Prove that this statement is false and write its negation.

Answer. We show that this statement is false and write its negation.

In order to show that the statement is false, it suffices to find $x \in B$, $y \in A$ with $2x - y > 1$ and $y^2 - x$ not a multiple of 3. For this purpose, take $x = 1$ and $y = -3$.

Negation of the statement. $(\exists x \in B)(\exists y \in A)((2x - y > 1) \wedge (y^2 - x \text{ is not a multiple of } 3))$.

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